Case Studies: Sports Analytics

UEFA Euro – Poisson, Skellam W/D/L and the draw problem

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Research Question

- Predict **win / draw / loss** for UEFA Euro 2028 matches from team covariates (GDP, market value, FIFA rank, UEFA points, Champions League players, group stage)
- Which model class and which tuning metric give the best **honest out-of-sample** W/D/L performance?
- In particular: can we predict **draws** at all?

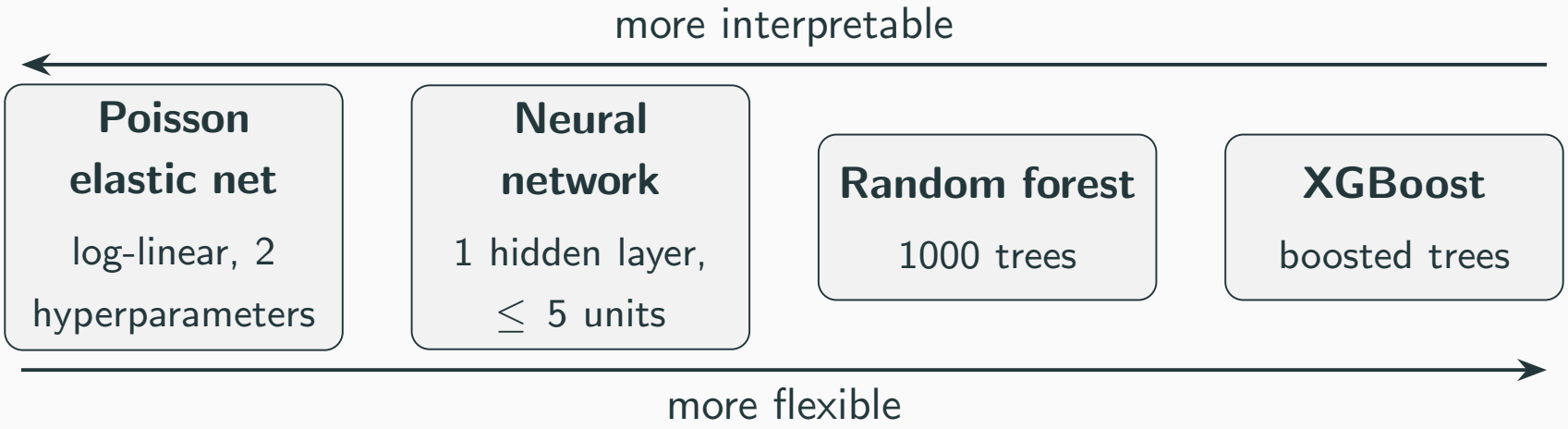
This week: Poisson elastic net in depth

- goals $\rightarrow$ Skellam $\rightarrow$ W/D/L $\rightarrow$ draw threshold
- evaluated with nested leave-one-Euro-out cross-validation

Assumptions

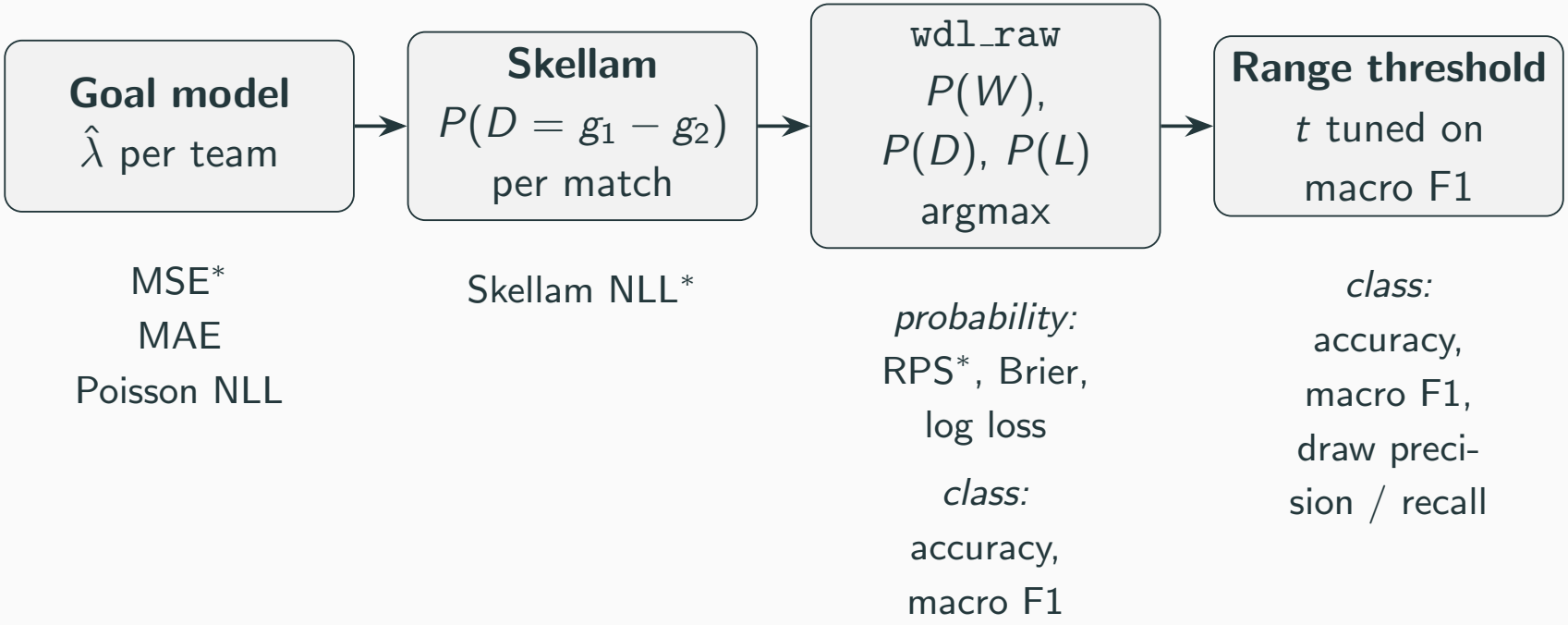
- **Goal is 2028, not 2024**: past Euros stand in for the future one
  - assumes dependence over time between Euros is negligible
  - e.g. squad market values have grown a lot since 2004; comparing the two teams of a match limits this, remaining drift is assumed negligible
- **Respect `match_id` in the split**
  - each match is two mirror-image rows (team vs opponent, covariates x and $-x$)
  - splitting them across train/test would leak the result
- **Respect tournament / year in the split**
  - a team's covariates (e.g. its 2024 market value) repeat in all its matches of that Euro
  - random row splits would let the model see the test tournament $\rightarrow$ optimistic scores

Model Class Choices



- All classes predict **goals per team**, then share the same Skellam W/D/L step (NN also tried direct classification)
- Same nested split structure for every class $\rightarrow$ comparable scores
- This week's deep dive: the Poisson elastic net

Choosing Evaluation Metrics



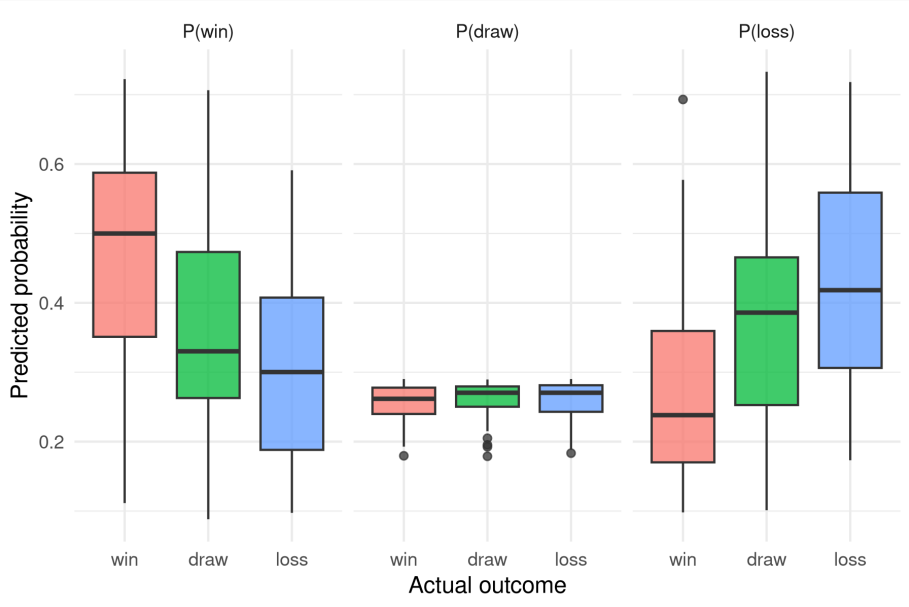
- All considered: macro F1, RPS, Poisson NLL, MSE, MAE, Skellam NLL
- * used as the hyperparameter **selection** metric (one experiment each)
- Probability metrics don't depend on t; class metrics are reported for argmax (`wdl_raw`) and for the tuned threshold

Draws Are Under-Predicted

Argmax W/D/L, nested outer predictions (selected on RPS)

	actual		
predicted	win	draw	loss
win	64	34	30
draw	0	0	0
loss	23	38	57

- 72 of 246 matches are draws (29%)
- argmax calls **0** draws
- max predicted $P(D) = 0.29$, mean 0.26
- historical-frequency baseline: also 0 draws



Approach to Fixing Draw Prediction

Range idea: a match is a toss-up when the three probabilities are close

$$\text{draw if } \max(P_W, P_D, P_L) - \min(P_W, P_D, P_L) < t, \text{ else argmax}$$

- Started with a pure, fixed threshold ($t = 0.15$)
- Then tuned t, first on **accuracy**: calling draws costs correct wins/losses, so accuracy barely rewards it
- Moved to **macro F1**: every class weighted equally, so getting draws right counts
- t chosen **leave-one-Euro-out**: 0.25 for 5 of 6 years, 0.225 for 2016; final rule for 2028: $t = 0.25$

Pooled outer predictions (selected on RPS)

t	draws	accuracy	macro F1
0	0	0.492	0.384
0.10	20	0.463	0.391
0.15	69	0.463	0.452
0.20	101	0.463	0.464
0.225	115	0.472	0.476
0.25	**121**	**0.476**	**0.480**
0.275	129	0.463	0.468
0.30	145	0.455	0.457

Draw Fix: Baseline vs Argmax vs Range-Adjusted

Nested outer predictions; historical-frequency baseline is leave-one-Euro-out

model	accuracy	macro F1	draws called	draw prec.	draw recall	draw F1
baseline	0.317	0.247	0	–	0.000	0.000
mse, argmax	0.492	0.384	0	–	0.000	0.000
mse, range	0.451	0.456	119	0.303	0.500	0.377
rps, argmax	0.492	0.384	0	–	0.000	0.000
rps, range	0.472	0.476	119	0.328	0.542	0.408
skel, argmax	0.492	0.384	0	–	0.000	0.000
skel, range	0.467	0.471	112	0.321	0.500	0.391

model	log loss	Brier	RPS
baseline	1.100	0.668	0.230
mse	0.999	0.600	0.196
rps	1.003	0.602	0.197
skel	1.000	0.601	0.197

- Probability metrics are identical for argmax and range-adjusted: only the decision rule changes
- Range adjustment: macro F1 0.38 $\rightarrow$ 0.48, draw recall 0 $\rightarrow$ 0.54, small accuracy cost

Group Dynamics

- **Communication:** [fill in]
- **Work splitting:** [fill in]
- **Meeting type:** [fill in]
- **Git repo:** [fill in]

Poisson Elastic Net 1

Hyperparameter Grid $7 \times 7 = 49$ combinations

$$\text{penalty} \in \{1.00e\text{-}10,\ 4.64e\text{-}09,\ 2.15e\text{-}07,\ 1.00e\text{-}05,\ 4.60e\text{-}04,\ 2.15e\text{-}02,\ 1.00\}$$
$$\times\ \text{mixture} \in \{0.000,\ 0.167,\ 0.333,\ 0.500,\ 0.666,\ 0.833,\ 1.000\}$$

Poisson Regression

$$\log(\mu(x)) = \beta_0 + \beta^T x$$

$$\ell(\beta|X, Y) = \sum_{i=1}^{N} \left[y_i(\beta_0 + \beta^T x_i) - e^{\beta_0 + \beta^T x_i} \right]$$

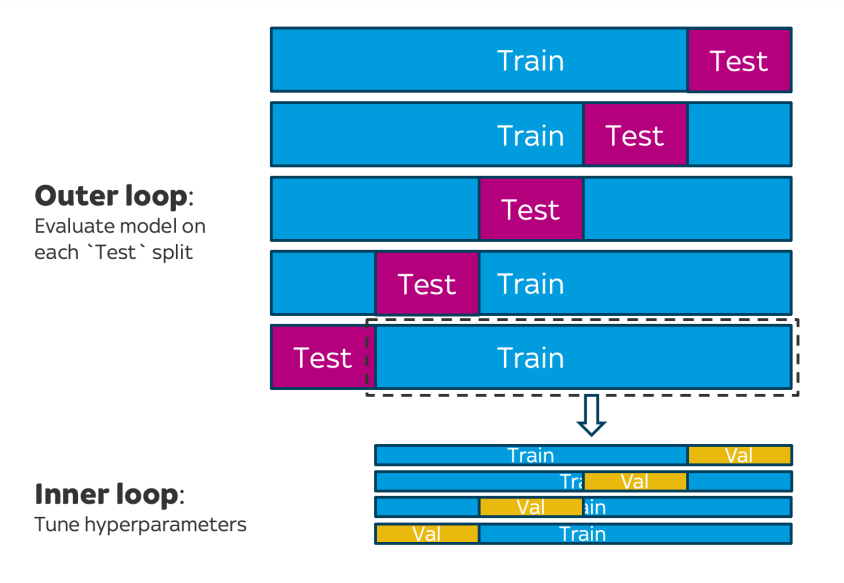
Penalized Objective

$$\hat{\beta} = \arg\min_{\beta_0, \beta} \quad -\frac{1}{N}\ell(\beta|X, Y) + \lambda \left[\frac{1-\alpha}{2} \sum_{j=1}^{p} \beta_j^2 + \alpha \sum_{j=1}^{p} |\beta_j| \right]$$

- **Penalty (λ):** Total amount of regularization.
- **Mixture (α):** Proportion of L1 regularization ($\alpha = 1$: pure lasso, $\alpha = 0$: ridge regression)

glmnet Poisson regression documentation link parsnip Poisson regression documentation link

Nested CV



- Test = 2020 (Outer 1)
  - Eval = 2024 (Inner 1.1)
  - Eval = 2008 (Inner 1.2)
  - ...
- Test = 2004 (Outer 2)
  - Eval = 2020 (Inner 2.1)
  - Eval = 2008 (Inner 2.2)
  - ...

Outer: Grouped Leave One Euro Out (6 Splits)

Inner: Grouped Leave One Euro Out (5 Splits x 6 Splits = 30)

Image Source: `https://towardsdatascience.com/two-common-pitfalls-to-avoid-when-doing-cross-validation-c68ed79c0e4e/`

Poisson: Skellam W/D/L vs Goal-Based Selection Metrics

Hyperparameter effects

Selected per outer fold

held out	mse λ / α	rps λ / α	skel λ / α
2004	0.0215 / 0.50	0.0215 / 1.00	0.0215 / 0.50
2008	0.0215 / 1.00	0.0215 / 1.00	0.0215 / 0.83
2012	0.0215 / 0.17	0.0215 / 1.00	0.0215 / 0.17
2016	0.0215 / 1.00	1 / 0.00	1e-10 / 0.00
2020	0.0215 / 1.00	0.0215 / 1.00	0.0215 / 0.67
2024	0.0215 / 1.00	1 / 0.00	0.0215 / 0.50

Final (2028) models

selected on	penalty λ	mixture α
mse	0.0215	1.0
rps	0.0215	1.0
skel	0.0215	0.5

Folds disagree, but mse and rps finals are identical; skellam differs only in mixture

Overall performance effects

Nested outer, pooled (se across 6 Euros)

metric	baseline	mse	rps	skellam
accuracy_raw	0.317 (.011)	0.492 (.033)	0.492 (.033)	0.492 (.033)
accuracy_t	0.280 (.028)	0.451 (.029)	0.472 (.026)	0.467 (.025)
brier	0.668 (.003)	0.600 (.010)	0.602 (.011)	0.601 (.010)
log_loss	1.100 (.004)	0.999 (.014)	1.003 (.016)	1.000 (.015)
macro_f1_raw	0.247 (.004)	0.384 (.025)	0.384 (.025)	0.384 (.025)
macro_f1_t	0.275 (.012)	0.456 (.027)	0.476 (.025)	0.471 (.024)
mae	0.870 (.023)	0.820 (.022)	0.819 (.022)	0.821 (.022)
poisson_log_loss	1.412 (.024)	1.349 (.022)	1.350 (.021)	1.349 (.021)
rmse	1.098 (.027)	1.027 (.023)	1.028 (.023)	1.028 (.022)
rps	0.230 (.005)	0.196 (.006)	0.197 (.006)	0.197 (.006)
skellam_nll	–	1.771 (.029)	1.774 (.029)	1.772 (.029)

_t: tuned range threshold, _raw: argmax. All differences between selection metrics are well within 1 se; every Poisson model clearly beats the baseline

Novel Covariate Finding Progress

Poisson Coefficient Interpretation (Final Model)

Selected on mse and on rps (identical model): $\lambda = 0.0215$, $\alpha = 1$ (lasso)

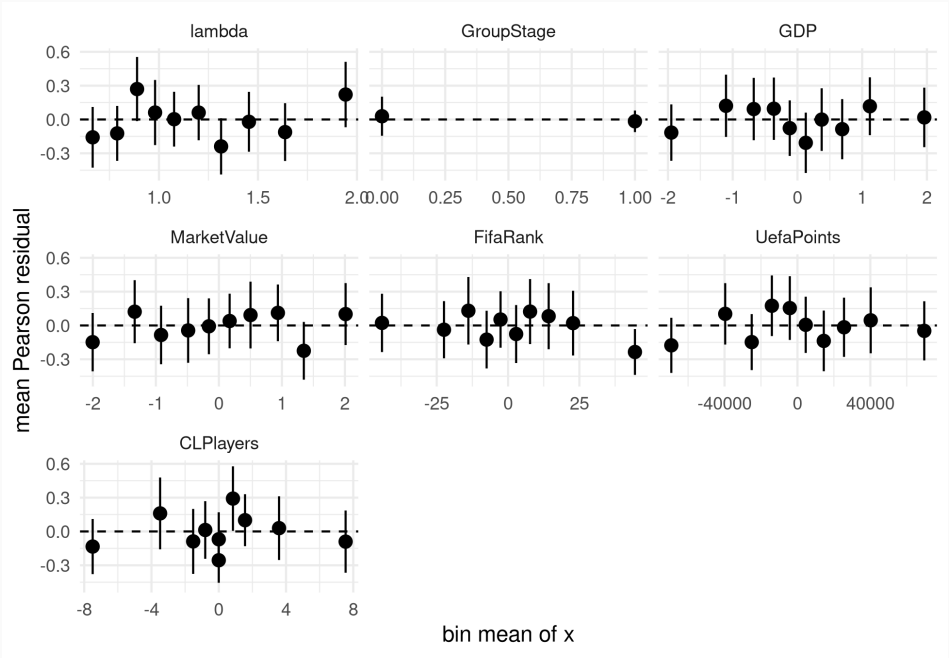
term	β_k	e^{β_k}	sd_x	$\beta_k\, sd_x$	$e^{\beta_k sd_x}$
(Intercept)	0.130	1.14	–	–	–
GroupStage	-0.00658	0.993	0.444	-0.00292	0.997
GDP	0.0958	1.10	1.09	0.105	1.11
MarketValue	0.157	1.17	1.20	0.188	1.21
FifaRank	-0.000402	1.00	24.4	-0.00981	0.990
UefaPoints	0	1.00	38800	0	1.00
CLPlayers	0.0262	1.03	3.99	0.105	1.11

- $\log \lambda = \beta_0 + \sum_k \beta_k x_k$: expected goals are **multiplied** by e^{β_k} per unit, $e^{\beta_k sd_x}$ per sd of x_k
- Team-minus-opponent differences: 1-sd market value advantage $\rightarrow \times 1.21$ expected goals ($+21\%$)
- UEFA points dropped by the lasso; strength covariates overlap, read them as a group
- Point estimates only (no standard errors from glmnet); lasso shrinks toward 0

Poisson Final Model Validation Checks

selected on	dispersion	match r	95% CI	p
mse	0.875	0.051	[-0.07, 0.18]	0.424
rps	0.875	0.051	[-0.07, 0.18]	0.424
skel	0.877	0.052	[-0.07, 0.18]	0.420

- **Dispersion** ≈ 0.88: no overdispersion
- **Within-match** $r = 0.05$, $p = 0.42$: Skellam's independence assumption holds
- **Log-linear form** (right): residuals scatter around 0
- $\Rightarrow$ draw problem is the W/D/L decision rule, not model misspecification



Next Steps

- Feature importance
- Potentially testing with an expanding window (train on past Euros only, predict the next)
- Standardizing data pipelines across model classes, to be sure we compare apples to apples

Backup: Final Model Selected on Skellam NLL

$\lambda = 0.0215$, $\alpha = 0.5$

term	β_k	e^{β_k}	sd_x	$\beta_k\, sd_x$	$e^{\beta_k sd_x}$
(Intercept)	0.142	1.15	–	–	–
GroupStage	-0.0272	0.973	0.444	-0.0121	0.988
GDP	0.102	1.11	1.09	0.111	1.12
MarketValue	0.153	1.17	1.20	0.183	1.20
FifaRank	-0.000735	0.999	24.4	-0.0179	0.982
UefaPoints	0	1.00	38800	0	1.00
CLPlayers	0.0280	1.03	3.99	0.111	1.12

Nearly identical to the mse / rps final model.